

Day 02

Data Storage & Cloud Services

Duration: 90 Minutes | Prepared by: Cyber.SoHo | 2026

Today's Agenda

LESSON 1

Data Storage

Storage architectures, data types, relational & NoSQL databases, data lakes & warehouses

LESSON 2

Cloud Services

Cloud models (IaaS/PaaS/SaaS), Big 3 providers, storage services, serverless, security

These two topics are the bedrock of every data science pipeline, every ML model, and every analytics dashboard.

LESSON 1

Data Storage

How data is persisted, organized, and scaled — from your smartphone to petabyte data centers

1.1 What Is Data Storage?

Recording and preserving digital information for future retrieval

Hardware layer — electromagnetic, optical, or electronic signals on physical media

Software layer — file systems, databases, object stores organizing signals into usable structures

Core operations — CRUD — Create, Read, Update, Delete

WHY IT MATTERS

Factor	Impact
Speed	10B-row query: seconds or hours?
Cost	\$50/mo vs. \$50,000/mo
Reliability	Data loss or self-healing?
Scalability	Breaks at 2× or auto-scales?

1.2 Three Fundamental Storage Architectures

Architecture	Analogy	Best For	Cloud Examples
Block Storage	Blank HD in fixed chunks	Databases, VMs, boot volumes	AWS EBS, Azure Disks
File Storage	Shared filing cabinet	Collaboration, home dirs	AWS EFS, Azure Files
Object Storage	Warehouse of labeled boxes	Data lakes, backups, ML data	AWS S3, Azure Blob, GCS

1.2 Architecture Performance Comparison

Metric	Block	File	Object
Latency	~1 ms	~5 ms	~20 ms
Max Throughput	1,000 MB/s	300 MB/s	5,000 MB/s
Cost (per GB/mo)	\$0.10	\$0.30	\$0.023
Max IOPS	64,000	35,000	5,500
Scale	Per-instance	Multi-user	Virtually unlimited

KEY TAKEAWAY

Block = fastest latency. Object = cheapest & most scalable. File = best for shared access. No single type does everything.

1.3 Three Data Categories

Structured ~20%	Unstructured ~80%	Semi-Structured varies
Rows & columns, predefined schema Examples: Bank transactions, employee records Storage: Relational DB / Columnar	No schema whatsoever Examples: Images, video, audio, PDFs, emails Storage: Object Storage (S3, Blob)	Tags/keys provide some organization Examples: JSON, XML, YAML, CSV, logs Storage: Document DB / Data Lake

KEY TAKEAWAY

Data structure dictates storage choice. Choosing wrong = 10× slower or 10× more expensive.

1.4 Relational Databases & SQL

Invented by Edgar F. Codd at IBM (1970)

Data organized into tables: rows = records, columns = attributes

SQL standardized by ANSI & ISO — most widely used query language

ACID Properties:

Atomicity — All or nothing

Consistency — Only valid states

Isolation — Concurrent transactions don't interfere

Durability — Committed data survives crashes

MAJOR RDBMS

Database	License	Strength
PostgreSQL	Open-source	Feature-rich
MySQL	Open-source	Widely deployed
SQL Server	Commercial	Enterprise
Oracle DB	Commercial	High-performance

KEY TAKEAWAY

Relational databases + SQL = foundation of structured data management. Master SQL — it is non-negotiable for data science.

1.5 NoSQL – Beyond Tables and Rows

"Not Only SQL" — Born from web-scale demands: millions of writes/sec, flexible schemas, horizontal scaling.

Type	Data Model	Best For	Examples
Document	JSON/BSON documents	Content mgmt, user profiles	MongoDB, CouchDB
Key-Value	Key → Value pairs	Caching, sessions, carts	Redis, DynamoDB
Column-Family	Dynamic column families	Time-series, IoT, analytics	Cassandra, HBase
Graph	Nodes & edges	Social networks, fraud detection	Neo4j, Neptune

1.5 The CAP Theorem

Any distributed database provides at most 2 of 3 guarantees simultaneously (Brewer, UC Berkeley, 2000):

C — Consistency

Every read returns the most recent write

A — Availability

Every request receives a response

P — Partition Tolerance

System works despite network splits

MongoDB → CP (Consistency + Partition) | Cassandra → AP (Availability + Partition)

KEY TAKEAWAY

NoSQL is not a replacement for SQL — it is a complement. Use the right tool for each workload.

1.6 Data Lakes vs. Data Warehouses

Feature	Data Warehouse	Data Lake
Data format	Structured, cleaned	Raw, any format
Loading pattern	ETL (Extract, Transform, Load)	ELT (Extract, Load, Transform)
Schema	Schema-on-write	Schema-on-read
Query speed	Very fast (optimized)	Slower (raw data)
Cost per GB	Higher	Lower
Analogy	Organized library	Warehouse of unopened boxes

1.6 Key Players & The Lakehouse

DATA WAREHOUSES

Snowflake • Amazon Redshift • Google BigQuery • Azure Synapse

DATA LAKES

AWS S3 + Glue • Azure Data Lake Storage • Google Cloud Storage

DATA LAKEHOUSE — The Best of Both Worlds

Combines low-cost lake storage + fast warehouse queries + ACID guarantees.
Key platforms: Databricks Delta Lake • Apache Iceberg • Apache Hudi

KEY TAKEAWAY

Lakes store everything cheaply. Warehouses query fast. Lakehouses combine both. Modern pipelines use all three.

LESSON 2

Cloud Services

How the cloud delivers storage, compute, and software as scalable, pay-per-use utilities

2.1 What Are Cloud Services?

Cloud computing = delivery of computing resources over the internet on a pay-as-you-go basis.
Instead of owning servers → rent them from a provider who manages everything.

NIST (National Institute of Standards and Technology) — 5 Essential Characteristics:

1. **On-demand self-service** — Provision resources automatically
2. **Broad network access** — Available via internet, browsers, APIs, CLIs
3. **Resource pooling** — Multi-tenant, dynamically assigned
4. **Rapid elasticity** — Scale up/down instantly to match demand
5. **Measured service** — Pay per use, monitored & reported

2.1 The Utility Model – Electricity vs. Cloud

Electricity Grid	Cloud Computing
Power plant	Data center
Electric grid	The internet
Wall outlet	API / web console
Meter on your house	Usage dashboard / billing
Pay per kWh	Pay per compute-hour / GB
No need to own a generator	No need to own servers

KEY TAKEAWAY

Cloud did for IT what the electric grid did for factories — transformed capital expense into a metered utility.

2.2 Cloud Service Models: IaaS, PaaS, SaaS

Model	You Manage	Provider Manages	Analogy
IaaS	OS, middleware, runtime, apps	Servers, storage, network	Empty apartment
PaaS	Apps and data only	Everything else	Furnished apartment
SaaS	Nothing (just use it)	Everything	Hotel stay

2.2 IaaS (Infrastructure as a Service)

Provides — Virtual machines, networking, storage

You manage — OS, patches, security, scaling

Best for — Custom workloads, GPU computing, regulatory control

Examples — AWS EC2, Azure VMs, Google Compute Engine

WHEN TO CHOOSE IaaS

Need full OS-level control

GPU/HPC workloads

Regulatory requirements

Experienced DevOps team

2.2 PaaS & SaaS

PaaS — Platform as a Service

Provides — Complete dev & deployment environment

You manage — Your code only

Best for — Web apps, APIs, microservices

Examples — Heroku, Google App Engine, Elastic Beanstalk

SaaS — Software as a Service

Provides — Fully functional apps via browser

You manage — Nothing — log in and use

Best for — Email, CRM, collaboration

Examples — Google Workspace, Microsoft 365, Salesforce

KEY TAKEAWAY

IaaS = max control & responsibility. PaaS = code & deploy. SaaS = just use it. Choose based on team skills.

2.2 Shared Responsibility Stack

Layer	On-Prem	IaaS	PaaS	SaaS
Applications	You	You	You	Provider
Data	You	You	You	Shared
Runtime	You	You	Provider	Provider
OS	You	You	Provider	Provider
Virtualization	You	Provider	Provider	Provider
Servers	You	Provider	Provider	Provider
Networking	You	Provider	Provider	Provider

2.3 The Big Three Cloud Providers

Dimension	AWS	Azure	GCP
Market Share	~31%	~25%	~11%
Launched	2006	2010	2008
Strength	Breadth of services	Enterprise / hybrid	Data & AI
Compute	EC2	Virtual Machines	Compute Engine
Object Storage	S3	Blob Storage	Cloud Storage
Data Warehouse	Redshift	Synapse	BigQuery
ML Platform	SageMaker	Azure ML	Vertex AI

2.3 How to Choose a Provider

Broadest Ecosystem

Most services, largest community, startup-friendly

→ **AWS**

Enterprise & Hybrid

Microsoft integration, Active Directory, SQL Server

→ **Azure**

Data & AI Leadership

BigQuery, TensorFlow, Kubernetes, competitive pricing

→ **GCP**

KEY TAKEAWAY

No universally 'best' provider — only the best fit. Most enterprises use 2+ providers (multi-cloud).

2.4 Cloud Storage Services

Provider	Object Storage	Block Storage	File Storage
AWS	S3	EBS	EFS
Azure	Blob Storage	Managed Disks	Azure Files
GCP	Cloud Storage	Persistent Disk	Filestore

2.4 Storage Tiering – Saving Money at Scale

Tier	Access Pattern	Price/GB/mo	Retrieval
Standard (Hot)	Frequent	\$0.023	Instant
Infrequent Access	Monthly	\$0.0125	Instant
Glacier (Cold)	Yearly	\$0.004	3–5 hours
Deep Archive	Rarely/Never	\$0.00099	12–48 hours

100 TB on Standard
\$2,300 / month

100 TB on Deep Archive
\$99 / month

KEY TAKEAWAY

Proper storage tiering saves 70–95% on costs. Always implement S3 Lifecycle Policies for aging data.

2.5 Serverless Computing

Write a function → define a trigger → provider handles everything else

Provider manages: provisioning, scaling (0 → millions), patching

You pay only for actual compute time (measured in milliseconds)

Scales to zero when idle — no idle cost

SERVERLESS ECONOMICS

~\$0.0000002 per 200ms exec

First 1M requests/mo = FREE

1M executions ≈ < \$1.00/mo

PLATFORMS

AWS Lambda

Azure Functions

Google Cloud Functions

KEY TAKEAWAY

Serverless = event-driven, pay-per-millisecond, zero-to-infinite scale. Ideal for pipelines, APIs, automation.

2.6 Cloud Security – Shared Responsibility

CUSTOMER RESPONSIBILITY

(Security IN the Cloud)

- Customer data & encryption config
- IAM policies & MFA
- Application security
- Network access controls & firewalls
- OS patching (IaaS)

PROVIDER RESPONSIBILITY

(Security OF the Cloud)

- Physical data centers & guards
- Networking hardware
- Hypervisor / host OS
- Storage drive destruction
- Power, cooling, fire suppression

2.6 Essential Security Concepts

Concept	Description
IAM (Identity & Access Mgmt)	Who can access what — Least Privilege principle
Encryption at Rest	AES-256 on stored data
Encryption in Transit	TLS (Transport Layer Security) for network data
VPC (Virtual Private Cloud)	Isolated network you control
MFA (Multi-Factor Auth)	Password + second verification factor

Key Compliance Frameworks:

SOC 2 (SaaS controls) • HIPAA (US health data) • GDPR (EU data protection) • ISO 27001 (info security)

KEY TAKEAWAY

A single misconfiguration can expose millions of records. The provider secures infrastructure; YOU secure your configs & data.

Lecture Summary

LESSON 1 — Data Storage

Block (fast), File (shared), Object (scalable + cheap)

Structured → SQL; Unstructured → Object Storage

NoSQL: flexible schemas, CAP theorem trade-offs

Lakes + Warehouses + Lakehouses = modern stack

LESSON 2 — Cloud Services

Cloud = computing as a utility (NIST 5 chars)

IaaS → PaaS → SaaS: less control, less work

Big Three: AWS (breadth), Azure (enterprise), GCP (AI)

Tiering saves 70–95%; Serverless = pay per ms

THE CONNECTION:

Data Storage is the WHAT. Cloud Services is the HOW. Together they form the bedrock of data science.

DAY 02 COMPLETE

Thank You

Data Storage & Cloud Services

Next Session: Day 03 — To Be Announced

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